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Section 5.4: Types of Databases

This section focuses on different types of databases, in particular, **flat-file** and **relational** databases.

We will discuss the benefits of using relational databases and how tables can be joined together to make entering and updating data more efficient .

Key Concepts of this section:

- # Understand the difference between **flat-file** and **relational databases**.
- # Understand the uses of **primary keys** and **foreign keys** when linking two database tables together.

The two different types of databases

There are two main types of computer databases:

1. **Flat-File**
2. **Relational**

1 Flat-File Databases

Flat-File databases hold all of their data in **one table only**.



Key Words:

One Table, Data

Redundancy.

They are only suitable for **very simple databases**.

The patient database is an example of a flat-file as all of the information is stored in one single table:

Flat-File (one table)

NOTE!

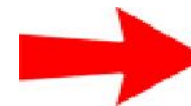
In your Section 11 exam (databases) you would only be expected to produce a flat-file.

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

Limitations of a Flat-File Database

The problems with using a flat-file databases are as follows:

- **Duplicated Data** is often **unnecessarily** entered.
- **Database space** is wasted with this duplicated data.
- **Duplicated Data** takes a long time to enter and update (unnecessarily).



**THESE PROBLEMS CAN BE
SUMMARISED WITH TWO
WORDS: DATA REDUNDANCY!**

What is Data Redundancy?

Data Redundancy is where you store the **same data many times** (duplicate data) in your table.

This repeated data needs to be **typed in over and over again** which takes a long time.

For example:-

The patients database contains several entries of duplicate data:

- **Doctor Id**
- **Dr Hyde**
- **Room 03**

NOTE!

Doctor Id 02, Dr Jekyll and Room 06 are also duplicates and therefore the descriptions here also apply to those data items.

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02	Dr Jekyll	06
198	Lisa	18-Dec-1979	Female	7498735	01	Dr Hyde	03
210	Frank	29-Apr-1983	Male	7943521	01	Dr Hyde	03
258	Rachel	8-Feb-1987	Female	8367242	02	Dr Jekyll	06

Duplicate

Duplicate

REMEMBER!

Data that is duplicated unnecessarily within a database is bad practice. If we had 100 patients who were all assigned Dr Hyde. His **Doctor Id, Name and Room Number** would have to be **entered 100 separate times**.

Also if Dr Hyde left the doctors surgery, we would have to update the new doctors details for every patient in the database.

What is the solution to Data Redundancy?

- # The way to avoid the data redundancy problems that come with flat-file databases is to create a **relational database**.

2 Relational Databases

- # Relational Databases use **two or more tables** linked together (to form a relationship).
- # Relational Databases do not store all the data in the same table.
- # Repeated data is moved into it's **own table** as shown in the image below:



Key Words:

Relationships,
Primary Key,
Foreign Key,
Common Field

Patient Table

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id
134	Jeff	4-Jul-1993	Male	7876453	01
178	David	8-Feb-1987	Male	8635467	02
198	Lisa	18-Dec-1979	Female	7498735	01
210	Frank	29-Apr-1983	Male	7943521	01
258	Rachel	8-Feb-1987	Female	8367242	02

Doctor Table

Doctor Id	Doctor	Room
01	Dr Hyde	03
02	Dr Jekyll	06

NOTE!

Here I have moved all the **repeating data** into a **table of its own**.

Now I have a **patient table** (for patient details) and a **doctor table** (for doctor details)

What is a relationship?

- # A relationship is formed when our **two tables are joined together**.

- # Relationships make use of **key fields** and **primary keys** to allow the **two tables to communicate** with each other and **share their data**.
- # Key fields are identified using a primary key as shown in the image below:

Key Field

Patient Id	Name	D.o.B	Gender	Phone	Doctor Id
134	Jeff	4-Jul-1993	Male	7876453	01
178	David	8-Feb-1987	Male	8635467	02
198	Lisa	18-Dec-1979	Female	7498735	01
210	Frank	29-Apr-1983	Male	7943521	01
258	Rachel	8-Feb-1987	Female	8367242	02

Key Field

Doctor Id	Doctor	Room
01	Dr Hyde	03
02	Dr Jekyll	06

NOTE!
Notice how I have only entered the **Doctor Name** and **Room Numbers** once.

REMEMBER!
The **key field** is used to hold **unique** (one of a kind) data.

- # Once the tables are linked together each one can read data from the other.
- # This means that we only need to enter the details of each doctor once instead of many separate entries.

NOTE!
When the link (relationship) is created, we will be able to **match each patient** with the correct **Doctor and Room Number** in the doctor table.

How do you form the relationship? (link the tables)

In order to link the tables we need to use a **common field**.

A common field is **data that appears in BOTH tables**.

If you look at the image below you will see that the common field in the patient database is **Doctor Id**:

Patient Table						Doctor Table		
Patient Id	Name	D.o.B	Gender	Phone	Doctor Id	Doctor Id	Doctor	Room
134	Jeff	4-Jul-1993	Male	7876453	01	01	Dr Hyde	03
178	David	8-Feb-1987	Male	8635467	02			
198	Lisa	18-Dec-1979	Female	7498735	01	02	Dr Jekyll	06
210	Frank	29-Apr-1983	Male	7943521	01			
258	Rachel	8-Feb-1987	Female	8367242	02			

NOTE!

The **primary key** (doctor Id) in the **doctor table** is being used to **link to the doctor Id** in the **patient table**.

At this stage you may have noticed the following:

- **Doctor Table** - The common field (Doctor Id) is **set as a key field**
- **Patient Table** - The common field is **not set as a key field**.

Common fields that are not being used as key fields are known as **foreign keys**.

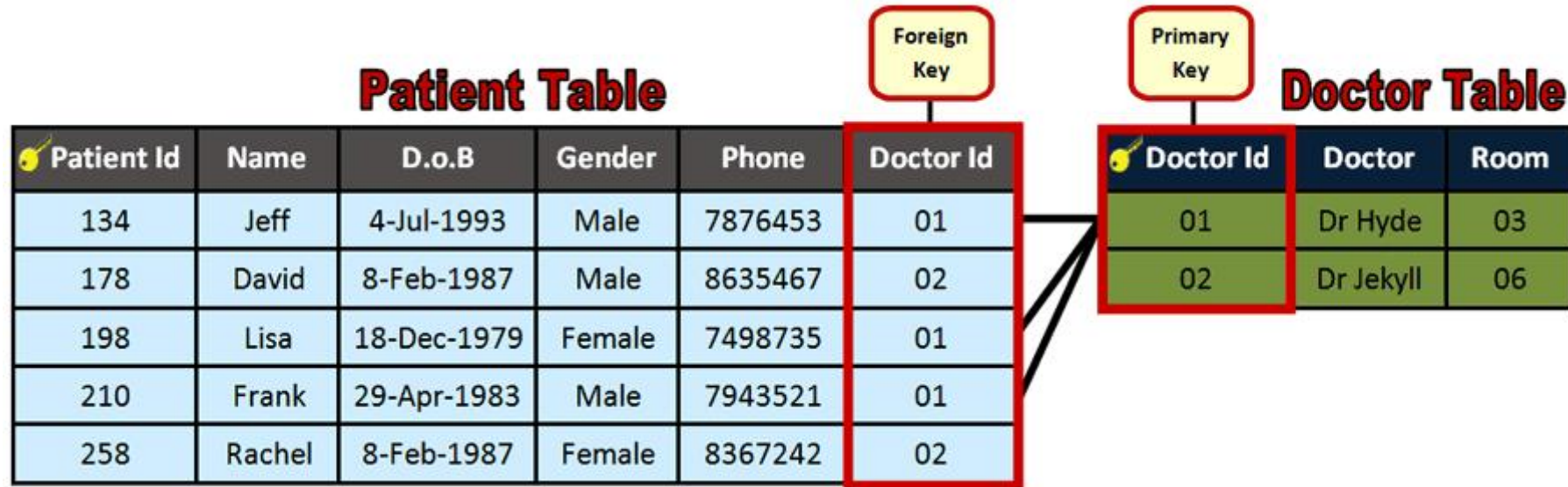
What is a foreign key and what are they used for?

A foreign key is a **regular field in one table** which is being used as the **key field in another table**.

Foreign keys are used to **provide the link** (relationship) between the tables.

For example:-

In our patient database, **Doctor Id** is a **key field in the Doctor Table** but is also being used in the **Patient Table as a foreign key**:



NOTE!

Without the **common field** (Doctor Id) and the use of **primary keys** and **foreign keys**, data cannot be shared between tables .

The foreign key (Doctor Id in the patient table) can then be used to match to the primary key (Doctor Id in the doctor table) and share the correct data.

For example:-

A patient with a **Doctor Id of 01** will be **automatically assigned** to **Doctor Hyde and Room 03**.

REMEMBER!

Now that we have linked our two tables we can update our doctor information very quickly.

For example:-

If Dr Hyde changed his **room number to 02**, we only need to **change this information once** in the doctor table. The new room would **automatically be assigned to every patient** who was under the care of Dr Hyde.

Benefits of using a Relational Database

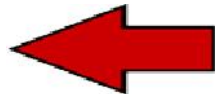
The advantages of using a relational database instead of a flat-file database are as follows:

- **Duplicated Data** is reduced.
- **Database space** is not wasted (due to unnecessary duplicated data).
- **Quicker to enter Data** as there are less duplicates.
- **Quicker to update Data**

Activity!

[Types of Databases - Task](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all tasks on the sheet.



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